

短视频观看诱发的颅内脑电研究

心理与行为科学系

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- 短视频已成为高普及、高时长占用的日常媒介情境

中国互联网络信息中心, 2026; 《中国网络视听发展研究报告》, 2026

- 个性化推荐、即时反馈与低操作成本, 使短视频观看具有持续强化特征。

Su et al., 2021; Hasan et al., 2018; Ferster & Skinner, 1957

- 杏仁核—海马系统可能参与情绪效价、主观偏好与凸显性信息的整合加工。

Pessoa & Adolphs, 2010; Hyman et al., 2006; Zheng et al., 2017

核心问题

杏仁核—海马系统
如何加工

情绪效价?

研究一

主观偏好?

研究二

刺激凸显性?

研究三

标准情绪刺激条件 (情绪效价)

标准情绪图片

效价判断

短视频刺激条件 (主观偏好)

短视频片段

固定时间窗
偏好评分

自然视频公开数据集 (凸显性信息)

Keles数据集

大规模
LFP + sEEG

研究一纳入被试

被试	年龄	性别	临床诊断	侧别	杏仁核电极	海马电极
Sub001	37	男	难治性癫痫	右侧	Amicro2	Amicro1、Bmicro1、C1、C2、C3
Sub004	19	男	难治性癫痫	右侧	A1、A2、A3、A4、A5	B1、B2、B3、B4、C2、C3、C4
Sub005	30	女	抑郁症	左侧、右侧	R-Amyg-MMa1、R-Amyg-MMa2、 R-Amyg-MMa3、L-Amyg-MMa1、 L-Amyg-MMa2、L-Amyg-MMa3、 L-Amyg-MMa4	R-hipp-BF1、R-hipp-BF2、L-Hip-BF1、L-Hip-BF2
Sub008	33	女	难治性癫痫	左侧	A-Amy-mTL1、A-Amy-mTL2、 A-Amy-mTL3、A-Amy-mTL4、 A-Amy-mTL5	B-aHippo1、B-aHippo2、C-cHippo1、C-cHippo2、C-cHippo3

情境判别范式

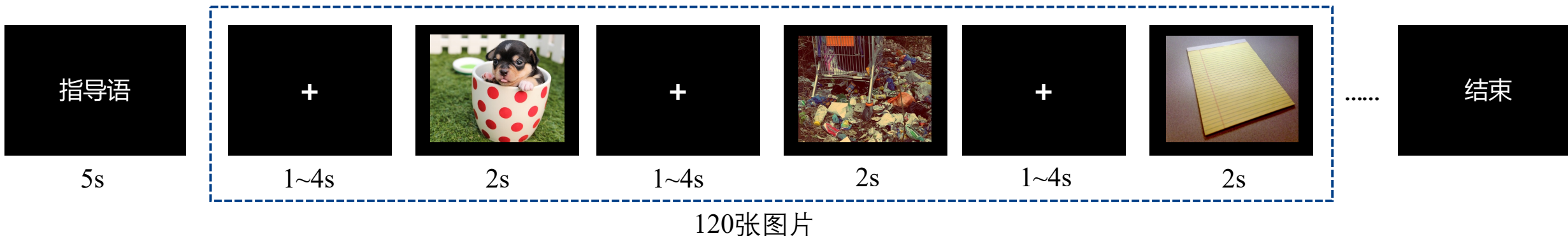
指导语:

在该任务中，您会看到一系列场景图片。图片呈现后，请您根据自己的真实感受，做出按键反应。请尽量**准确**地反应，不必追求速度。

如果您看到图片后产生了难过、厌恶、不安、愤怒等消极的情绪，按“1”键；

如果图片没有使您产生明确的情绪变化，按“2”键；

如果您看到图片后产生了愉悦、轻松、满足、开心等积极的情绪，按“3”键。



➤ 电极重建

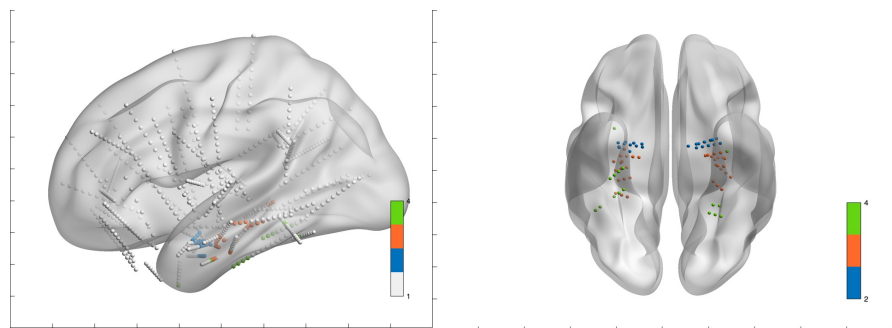
1. 术后CT/MRI造影
2. 配准并叠加到原始解剖空间中
3. 识别电极轴上的每个宏电极和微电极位点
4. 将原始空间映射到MNI空间
5. 结合临床Desikan-Killiany标签完成QC

➤ LFP 预处理

1. 降采样到1000Hz
2. 滤波: 1-200Hz带通滤波, 49-51Hz陷波去工频
3. 对齐行为数据, 合并事件
4. QC: 对两种策略筛选后的试次取交集

➤ 时频分析

1. 小波变换: 选取 $[-0.5, 0]$ s为基线窗, $[0, 2]$ s为分析窗, 分析2-100 Hz的事件相关功率变化
2. 以未校正 $p < 0.05$ 作为成簇阈值, 基于符号翻转置换方法进行1000次簇置换, 置换零分布中最大簇统计量大于或等于该真实簇统计量的比例 < 0.05 视作显著簇



Robust规则: 用基线窗和响应窗计算RMS比值和峰值, 在各条件内做robust z-score, 剔除 $z > 4$, 至少命中5个通道的试次

幅度规则: 在全epoch内检测 $|signal| \geq 200\mu V$, 并按条件分组做峰值 $\text{mean} \pm 3\sigma$ 判定, 剔除至少命中3个通道的试次

$$\psi_f(t) = \frac{1}{\sqrt{\sigma_f} \sqrt{\pi}} \exp(2\pi i f t) \exp\left(-\frac{t^2}{2\sigma_f^2}\right)$$

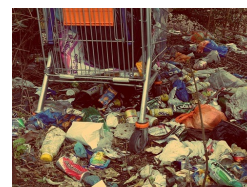
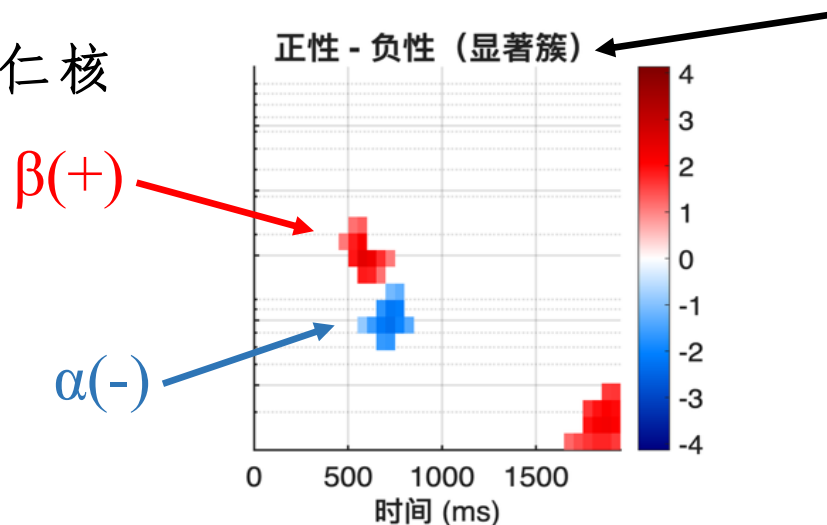
$$W_x(t, f) = \int_{-\infty}^{+\infty} x(\tau) \psi_f^*(\tau - t) d\tau$$

$$P(t, f) = |W_x(t, f)|^2$$

$$P_{dB}(t, f) = 10 \log_{10} \left(\frac{P(t, f)}{\frac{1}{|B|} \sum_{\tau \in B} P(\tau, f)} \right)$$

$$\Delta P_{c_1-c_2}(t, f) = \bar{P}_{c_1, dB}(t, f) - \bar{P}_{c_2, dB}(t, f)$$

➤ 杏仁核

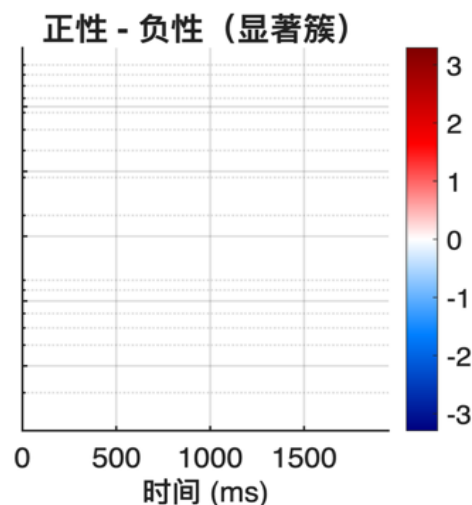


主簇：刺激后450-800 ms、5.8-20.2 Hz

副簇：刺激后1650-1950 ms、2.0-4.1 Hz

在刺激呈现后约500-800 ms的时间窗内，正性图片相较负性图片诱发了更明显的 α 频段功率减弱和 β 频段功率增强

➤ 海马



未观察到与杏仁核同样清晰的效价相关调制

标准化情绪图片能够在杏仁核中诱发频段特异性的效价加工信号

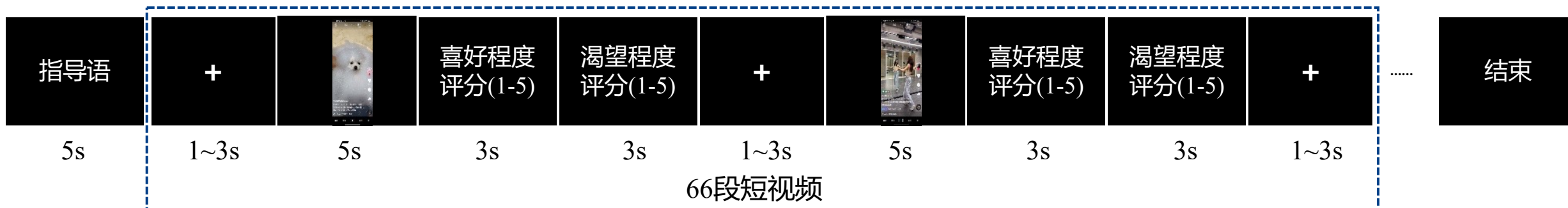
研究二纳入被试

被试	年龄	性别	临床诊断	侧别	杏仁核电极	海马电极
Sub001	37	男	难治性癫痫	右侧	Amicro2	Amicro1、Bmicro1、C1、C2、C3
Sub004	19	男	难治性癫痫	右侧	A1、A2、A3、A4、A5	B1、B2、B3、B4、C2、C3、C4
Sub008	33	女	难治性癫痫	左侧	A-Amy-mTL1、A-Amy-mTL2、 A-Amy-mTL3、A-Amy-mTL4、 A-Amy-mTL5	B-aHippo1、B-aHippo2、C-cHippo1、C-cHippo2、C-cHippo3

短视频观看范式 (N=3)

指导语:

在该任务中，您将观看一系列短视频的开头五秒。每段视频看完后，您都要先后对喜好程度和继续观看该视频的渴望程度进行1-5评分。



- 自有视频库来自抖音新注册用户的随机推送
- 定义主观评分 ≥ 4 为喜欢， ≤ 2 为不喜欢
- 喜好和渴望高度相关 (Spearman's $r = 0.85, p < 0.0001$)，故仅考察喜好

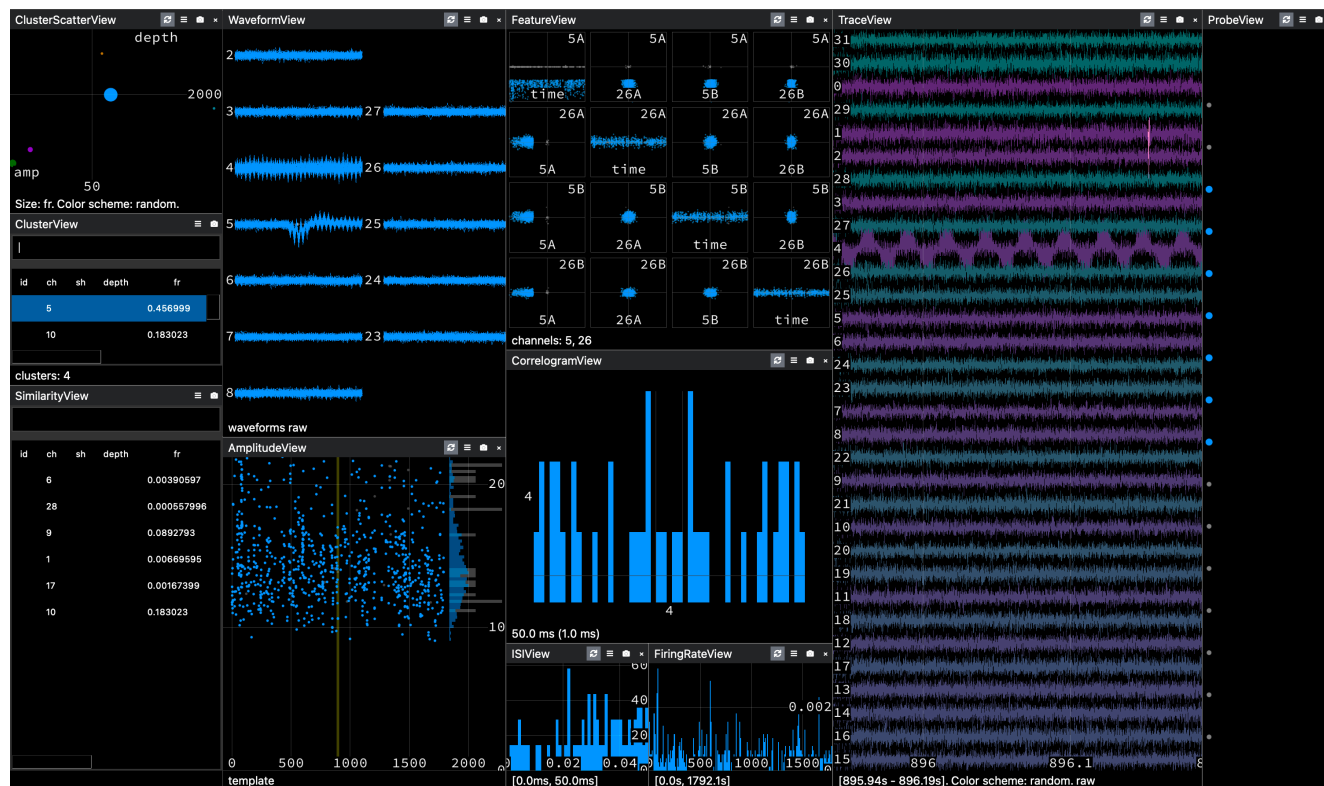
➤ spike预处理（独立于LFP采集）

1. 转换数据格式至二进制
2. 使用kilosort完成spike sorting
3. 基于phy筛选、合并、重组神经元并QC
4. 将每个放电单元的spike time对齐事件窗，
计算高斯核平滑瞬时放电率（IFR）

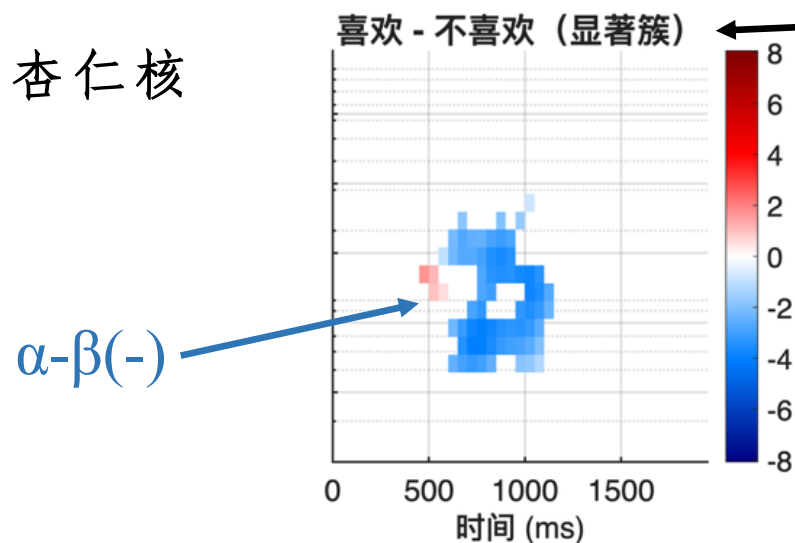
$$r_u(t) = \sum_{n=1}^{N_u} K_{\sigma}(t - s_{u,n})$$

$$K_{\sigma}(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

$$\sigma = 50 \text{ ms}, \quad \Delta t = 10 \text{ ms}$$

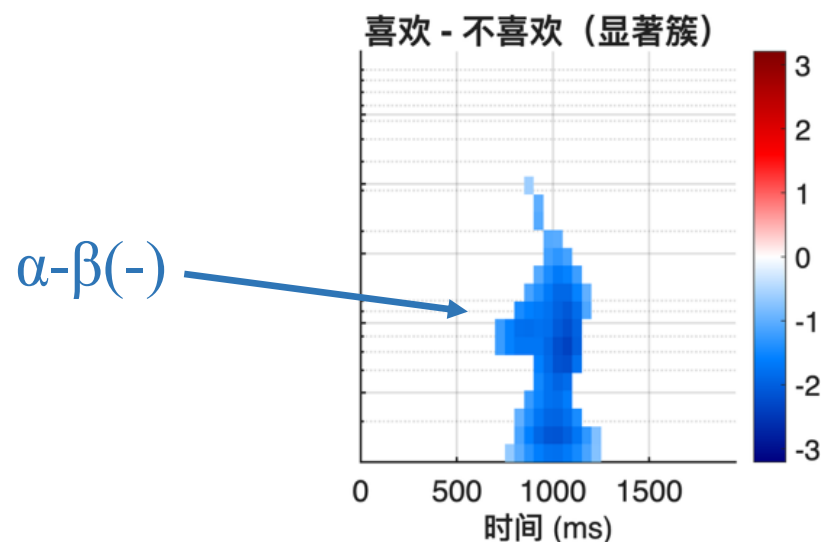


➤ 杏仁核



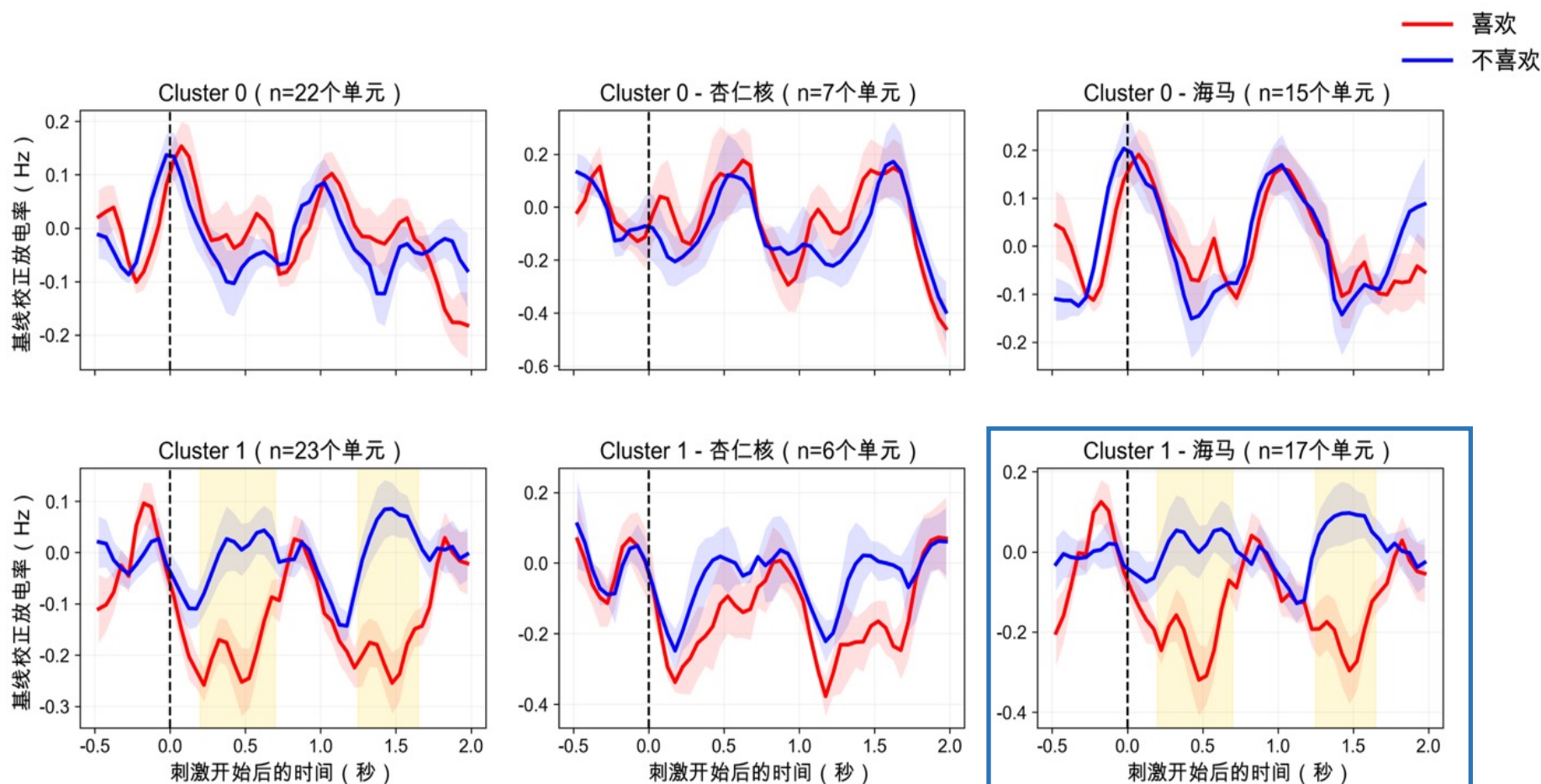
刺激后450-1100 ms、4.9-24.2 Hz, 喜欢的视频诱发了更明显的 α 频段和 β 频段功率减弱

➤ 海马



刺激后700-1200 ms、2.0-28.9 Hz, 喜欢的视频诱发了更明显的 α 频段和 β 频段功率减弱

与研究一相比, 除了杏仁核的参与, 海马在短视频条件下也表现出明显的偏好相关活动



对各单元的基线校正IFR反应曲线进行**K-means**聚类，得到两个主要响应模式簇：

Cluster 0: 22个单元，
杏仁核 (7)、海马 (15)

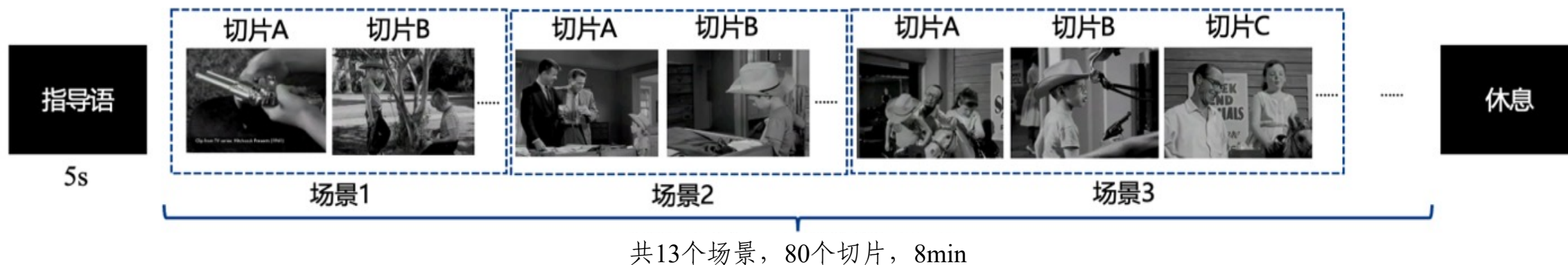
Cluster 1: 23个单元，
杏仁核 (6)、海马 (17)

杏仁核单元中两个簇的喜欢与不喜欢条件差异均不显著，上述显著差异主要来源于**海马**单元

海马放电活动可能是该簇水平**偏好效应**的主要贡献来源

Keles数据集（N=16，session=27）（Keles et al., 2024）

连续观看希区柯克导演的黑白电影*Bang! You're Dead*，将原始30分钟的内容剪辑为8分钟。



凸显性：刺激因其强度、新异性、情绪意义或任务相关性而从背景信息中突显出来，并优先捕获注意和引发神经加工的特性



➤ 功能连接和方向性分析

在杏仁核和海马电极之间计算不同频段的功能连接强度

使用谱格兰杰因果分析 (sGC) 估计两个脑区之间的方向性信息流

右式中 $S_{xy}(f)$ 为交叉谱, $C_{xy}(f)$ 为相干性, p 为MVAR模型阶数, A_k 为自回归系数矩阵, Σ 为残差协方差, $D(f)$ 为海马到杏仁核相对于反向的信息流差异

➤ spike分析

在研究二spike分析方法的基础上, 进一步提取70-150 Hz高 γ 包络, 并在每个试次内计算IFR与高伽马包络之间的Spearman相关, 分别得到脑区内耦合和跨脑区耦合指标。

为描述杏仁核与海马放电率变化的相对时间关系, 对每个试次的杏仁核IFR和海马IFR进行标准化互相关分析, 在 ± 300 ms 范围内寻找峰值相关对应的时滞; 正值表示杏仁核领先海马, 负值表示海马领先杏仁核。

$$C_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)}$$

$$X_t = \sum_{k=1}^p A_k X_{t-k} + \varepsilon_t$$

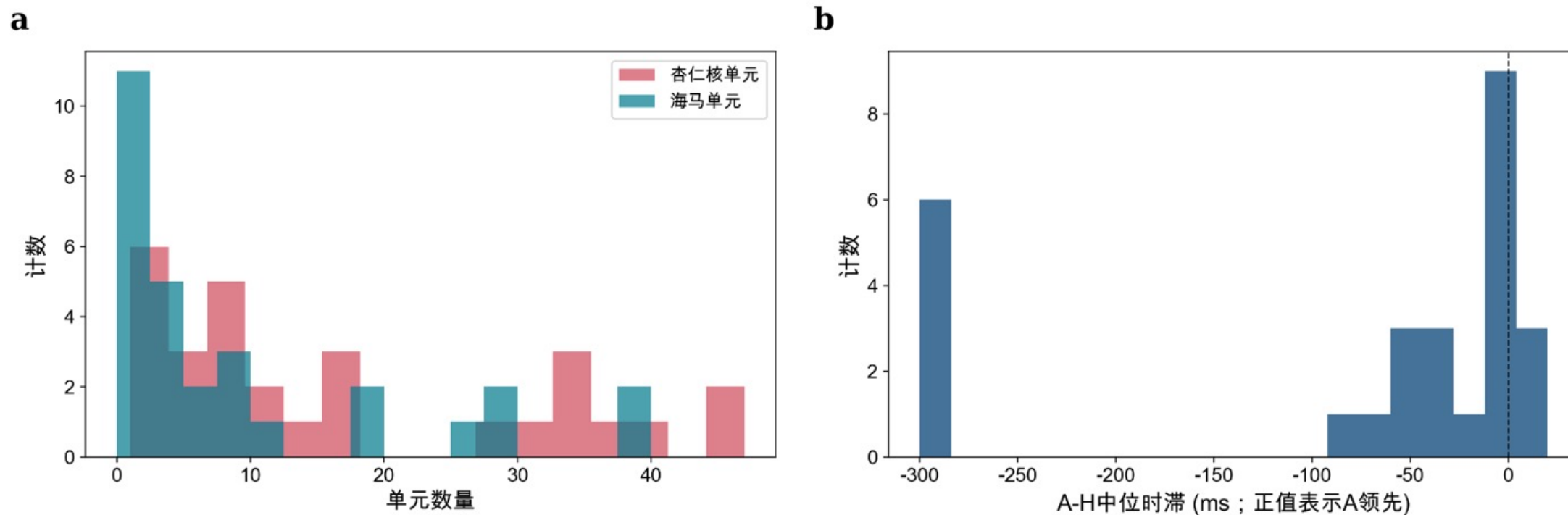
$$S(f) = H(f)\Sigma H^*(f)$$

$$H(f) = \left(I - \sum_{k=1}^p A_k e^{-i2\pi f k} \right)^{-1}$$

$$F_{(x \rightarrow y)}(f) = \ln \left(S_{yy}(f) / S_{(yy|x)}(f) \right)$$

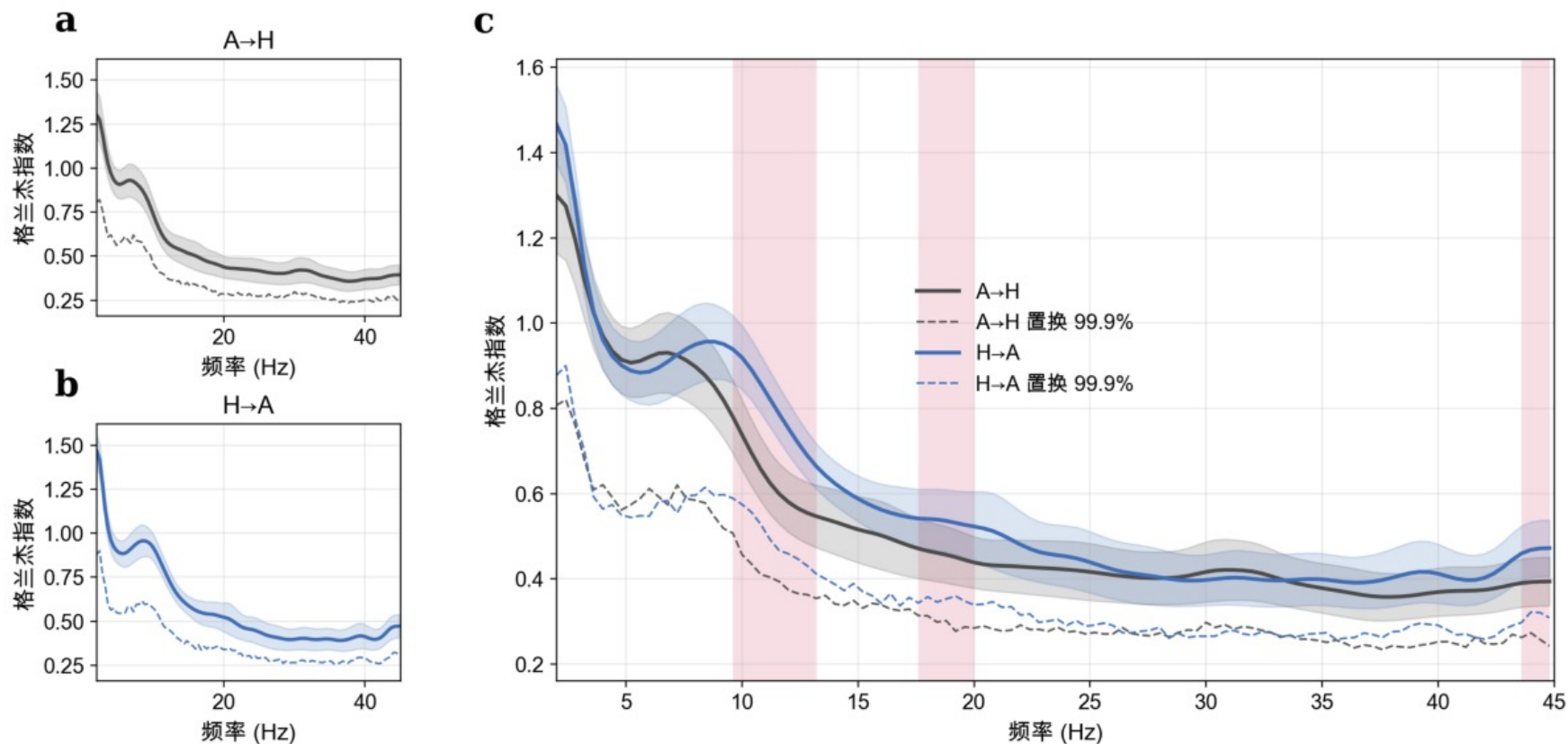
$$F_{(y \rightarrow x)}(f) = \ln \left(S_{xx}(f) / S_{(xx|y)}(f) \right)$$

$$D(f) = F_{\text{HPC} \rightarrow \text{AMY}}(f) - F_{\text{AMY} \rightarrow \text{HPC}}(f)$$



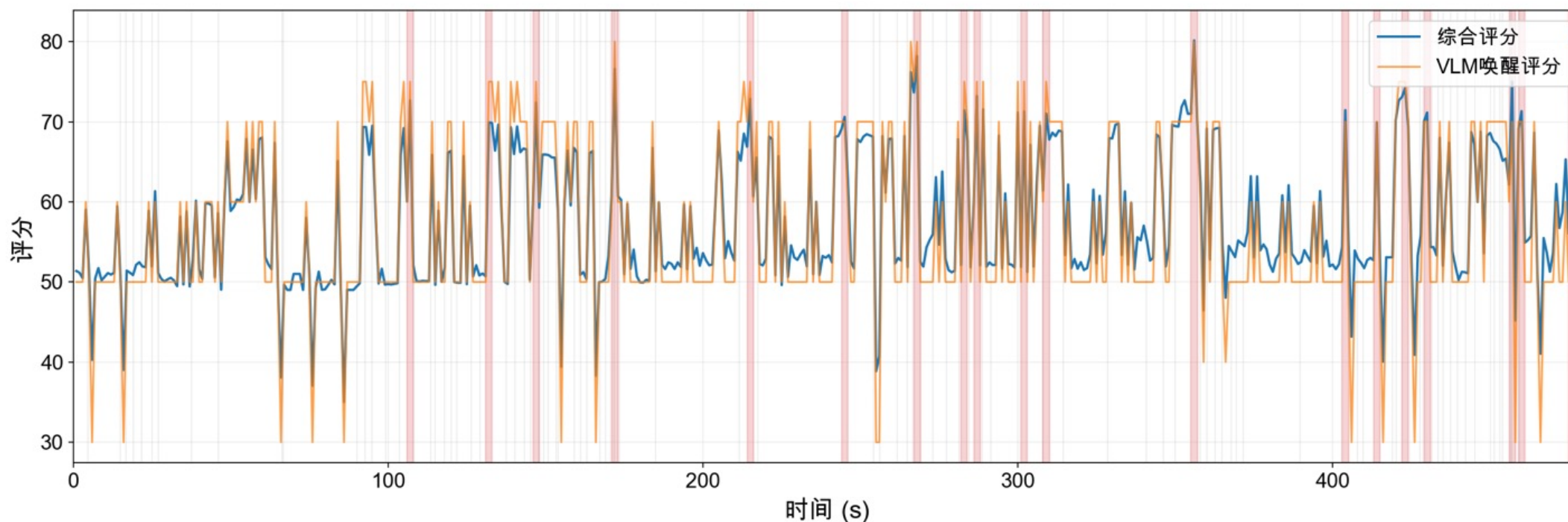
负时滞记录段的比例显著高于随机水平 (16/19, 双侧 $p = 0.004$)

海马对事件相关放电变化的时间响应总体上早于杏仁核



组水平平均sGC在A→H方向为0.535，H→A方向为0.584，平均方向差为0.049；
FDR校正后显著频点主要分布在9.6-13.2 Hz、17.6-20.0 Hz和43.6-44.8 Hz三个频段

涉及场景切换时，海马到杏仁核方向的信息流可能更强



使用VLM模型Qwen/Qwen2.5-VL-3B-Instruct和声学特征共同评价视频材料的显著性水平，
从477个候选窗口中选出19个高凸显性窗口

模型输出 *ArousalScore* 和 *Confidence*

按照 $AudioSalience = 0.6 \times RmsEnergy + 0.4 \times SpectralFlux$ 计算声学显著性，0-100归一化后综合得到

$$FinalScore = 0.7 \times ArousalScore + 0.2 \times Confidence + 0.1 \times NormalizedAudioSalience$$

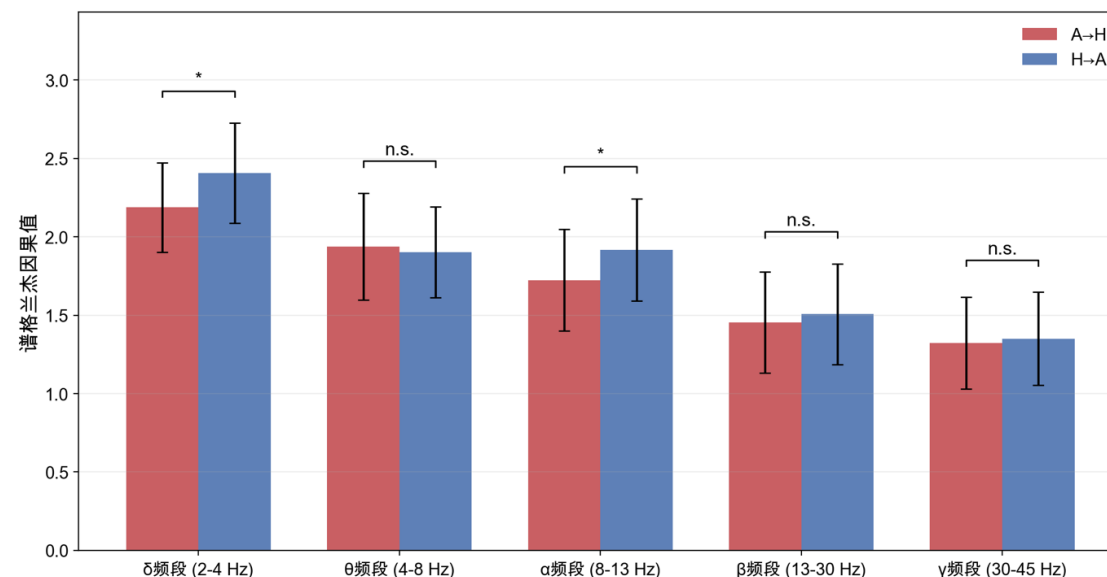
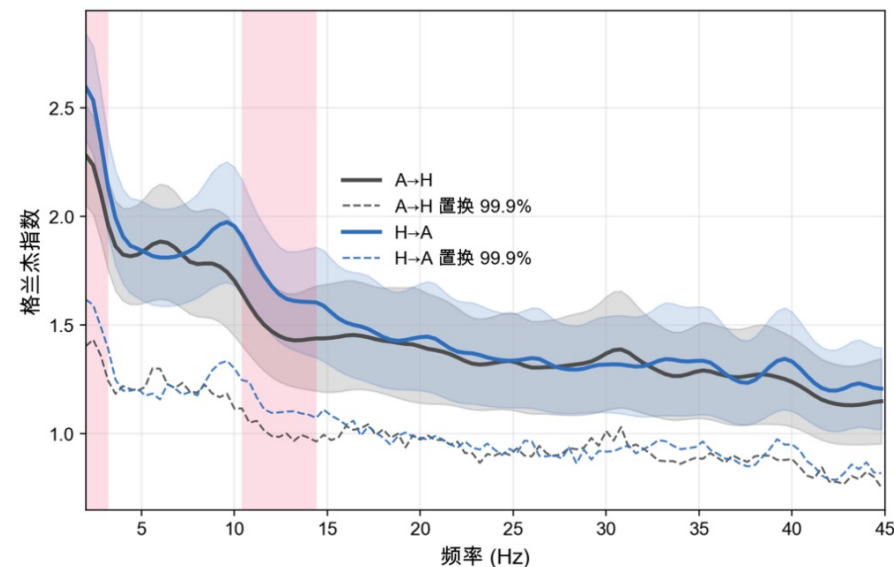
定义 $FinalScore \geq 70$ 为高凸显性窗口



使用VLM标注事件 $H \rightarrow A$ 平均sGC为1.501， $A \rightarrow H$ 平均sGC为1.438，方向差数值上仍表现为 **$H \rightarrow A$ 更强**

频段	频率范围	$A \rightarrow H$ 均值	$H \rightarrow A$ 均值	统计结果	p值
δ	2–4 Hz	2.187	2.406	$H \rightarrow A > A \rightarrow H$	.011
θ	4–8 Hz	1.937	1.902	n.s.	.730
α	8–13 Hz	1.723	1.917	$H \rightarrow A > A \rightarrow H$	.003
β	13–30 Hz	1.453	1.507	n.s.	.095
γ	30–45 Hz	1.323	1.351	n.s.	.148

高唤醒片段锁定条件下的方向性差异主要表现为 **δ 和 α 频段的海马→杏仁核优势**



主要结论

杏仁核—海马系统
如何加工

情绪效价?

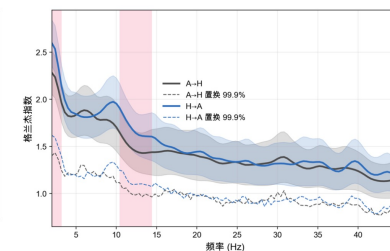
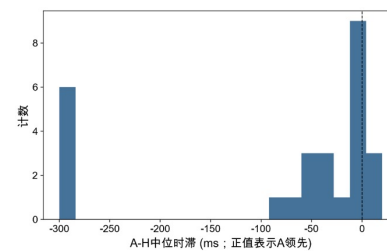
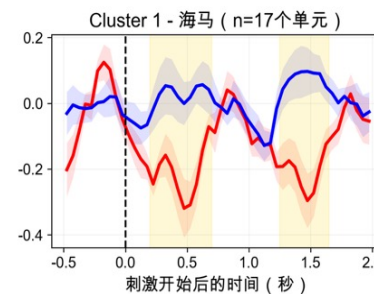
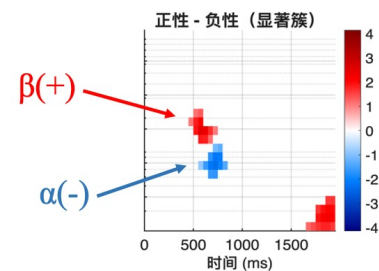
杏仁核: θ/α 至 β 频段活动改变
海马: 无显著变化

主观偏好?

喜欢视频放电率降低
主要来自海马

刺激凸显性?

海马放电比杏仁核更早
海马 → 杏仁核信息流更强



局限与展望

1. 样本量有限, 缺乏来自健康被试的直接证据; spike分析中的边缘系统单元数量也有限
2. 尚未系统纳入伏隔核、眶额皮层、前扣带回等与奖赏、控制和决策相关的区域

中国互联网信息中心 (2026). 第 57 次中国互联网络发展状况统计报告.

<https://www.cnnic.cn/n4/2026/0304/c88-11549.html>

《中国网络视听发展研究报告 (2026)》. 第十三届中国网络视听大会

<https://www.cnnic.cn/n4/2026/0417/c326-11590.html>

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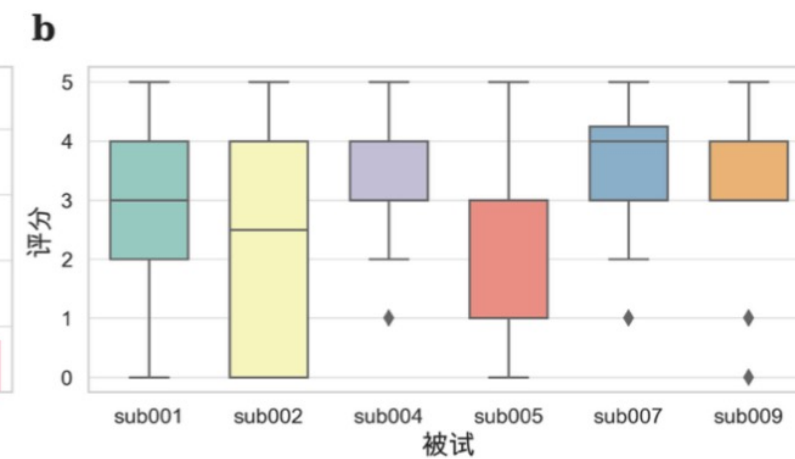
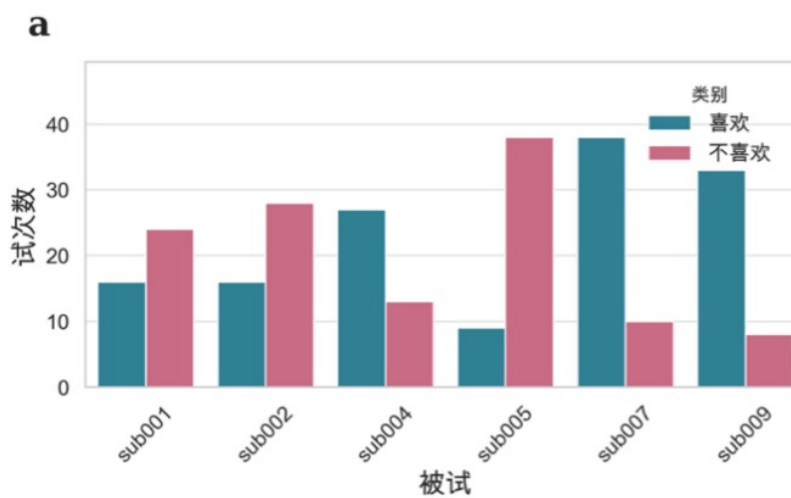
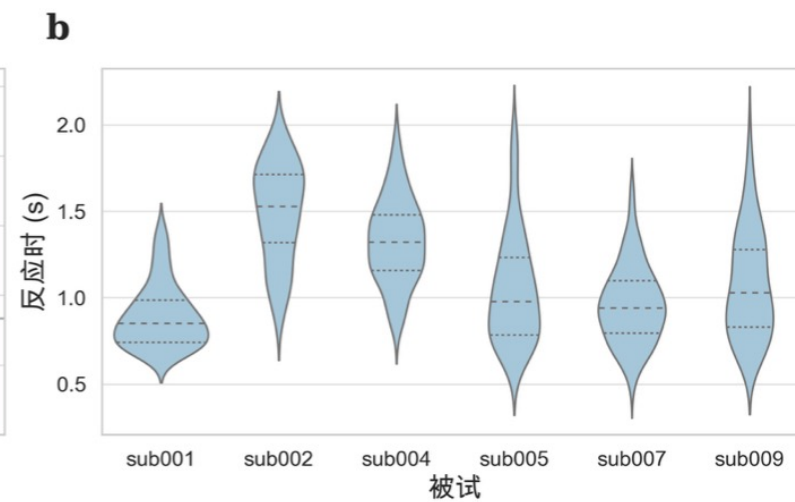
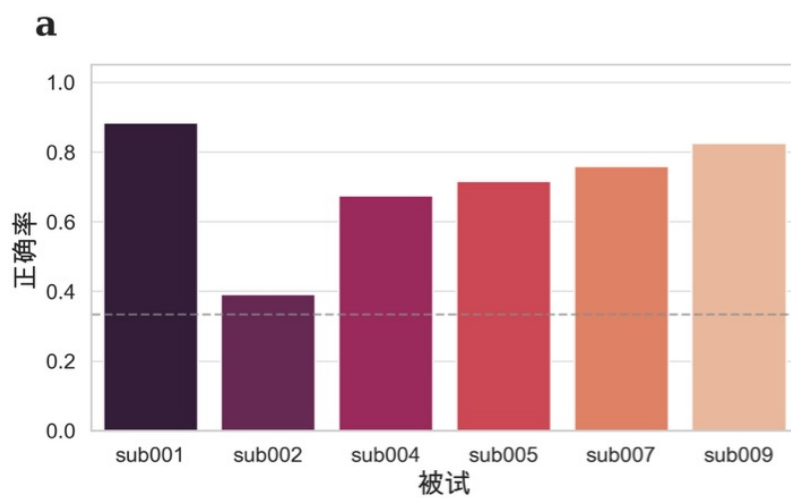
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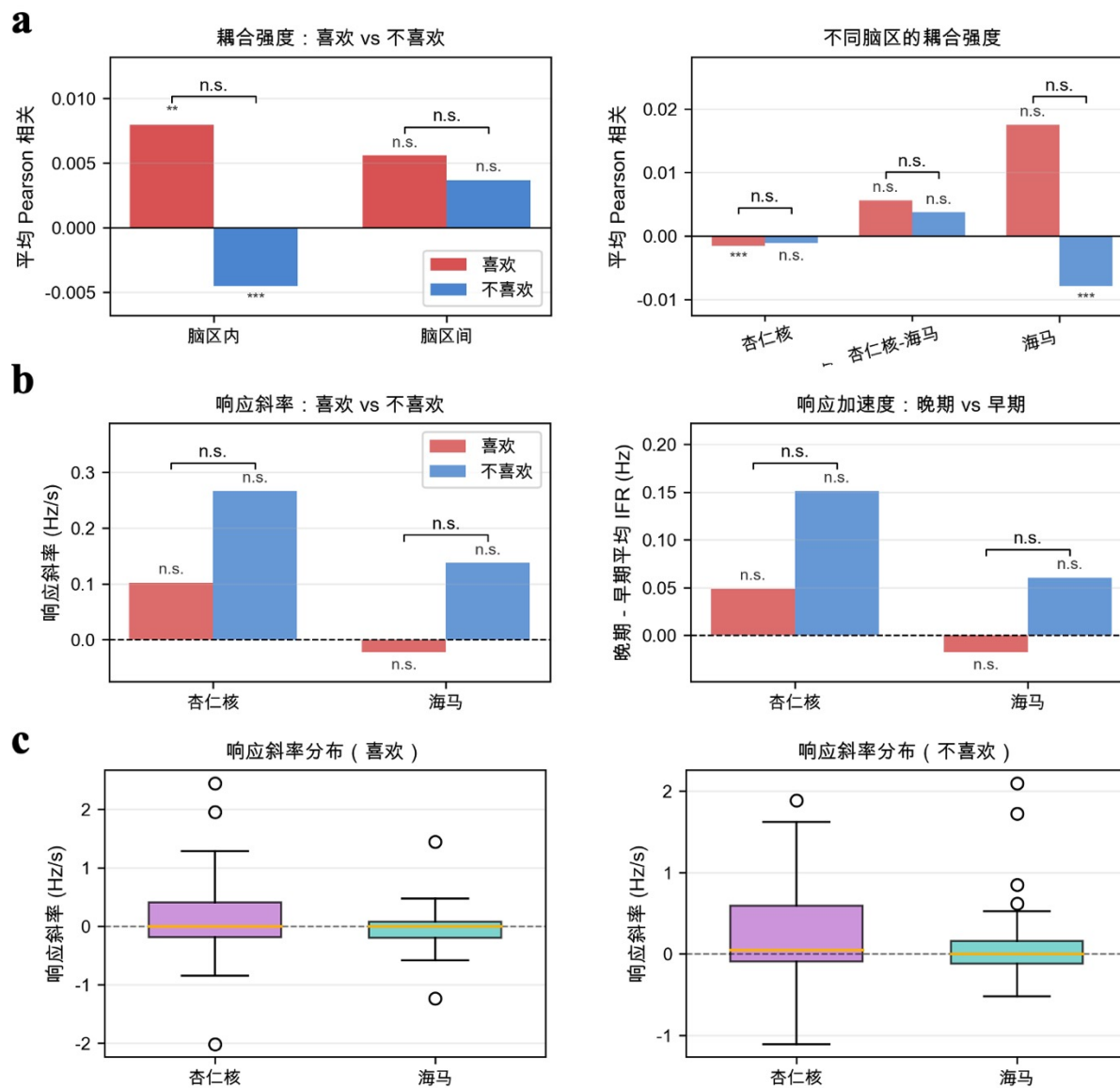
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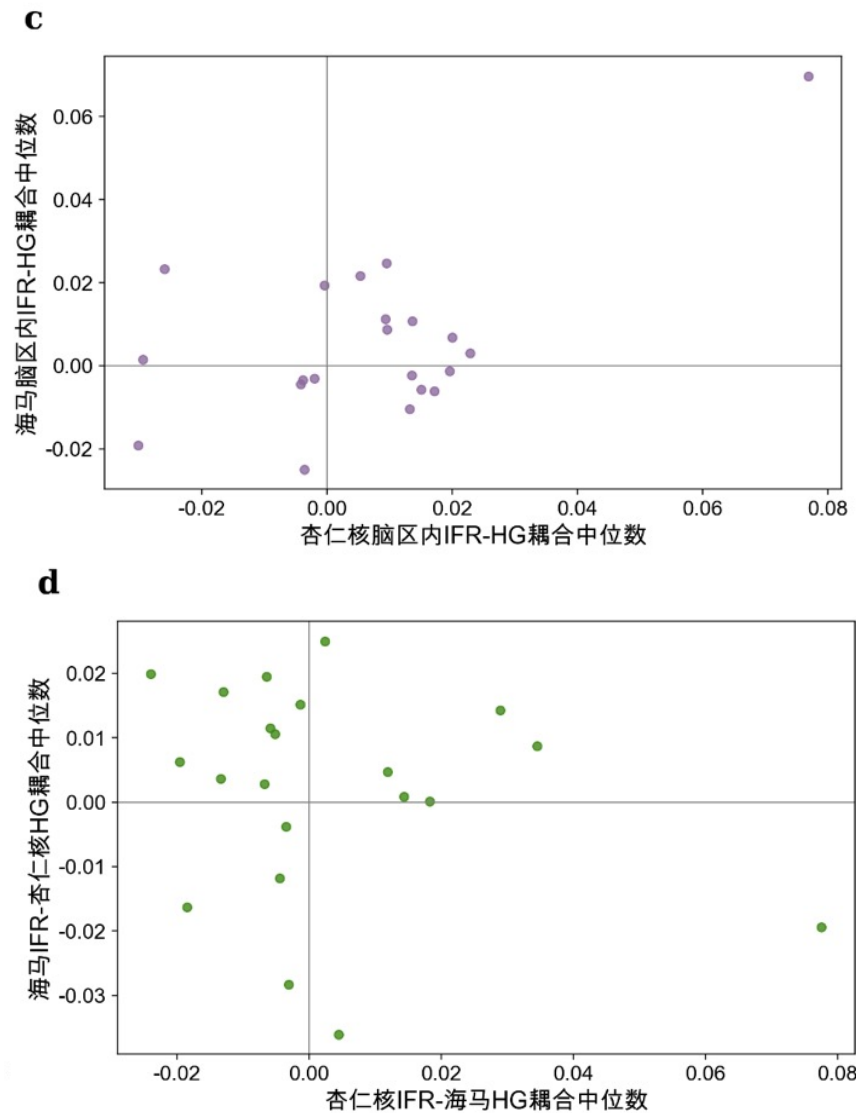
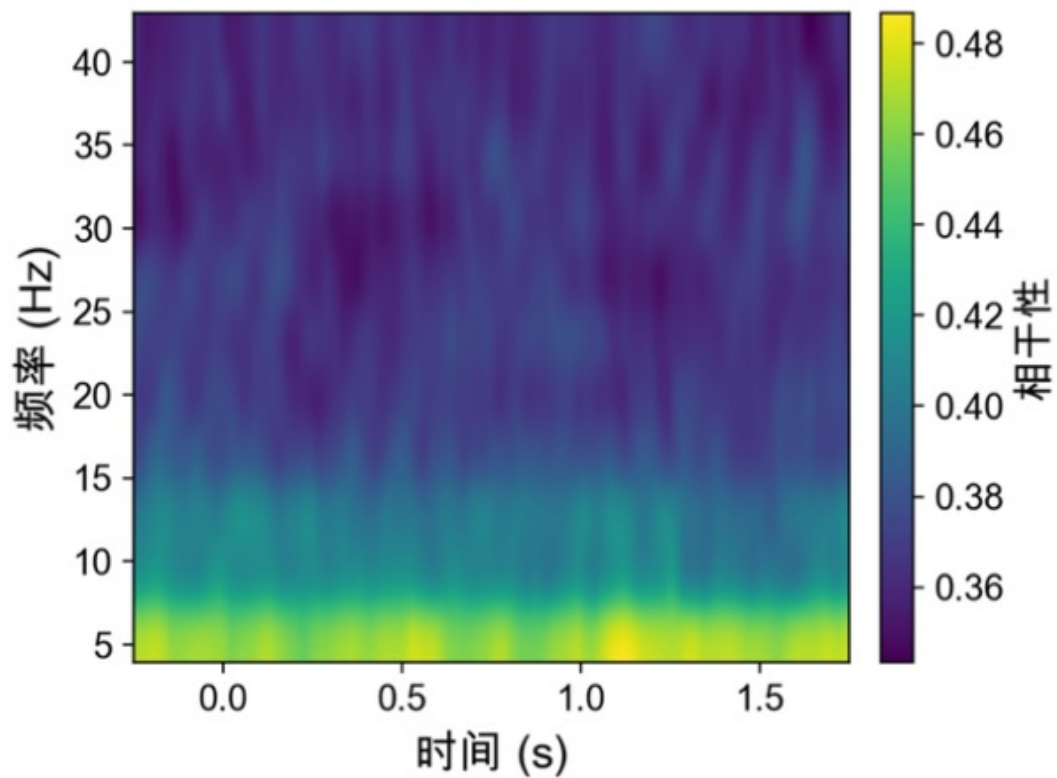
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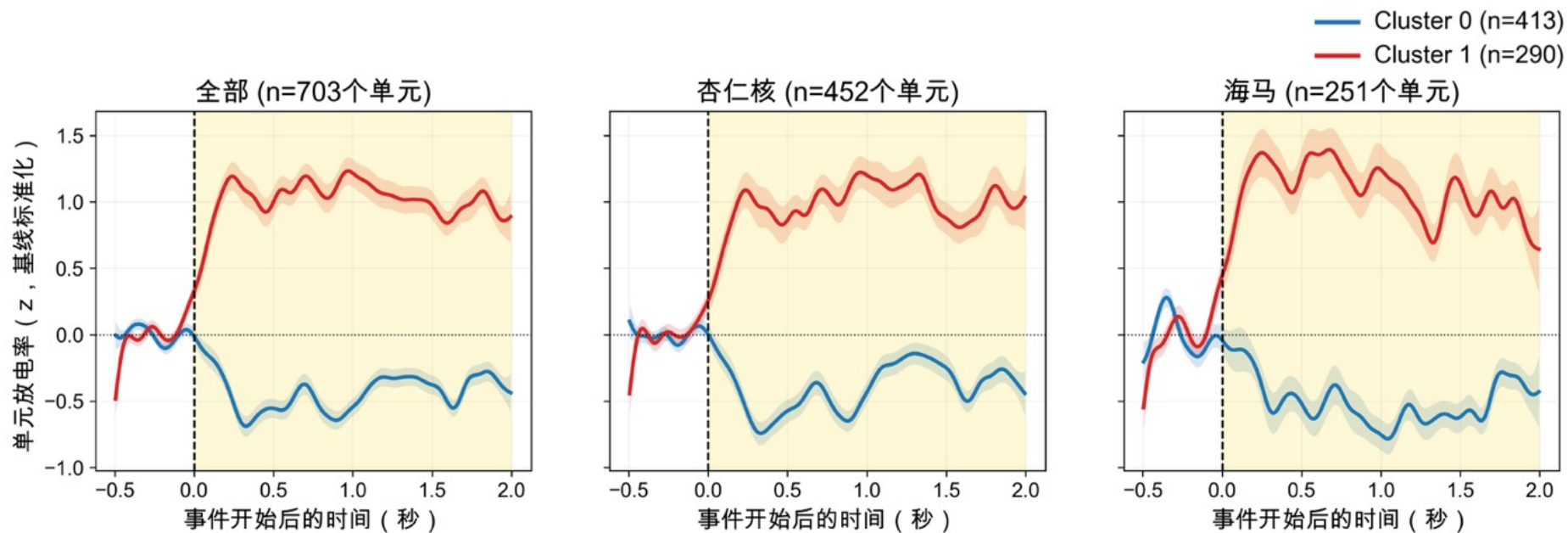
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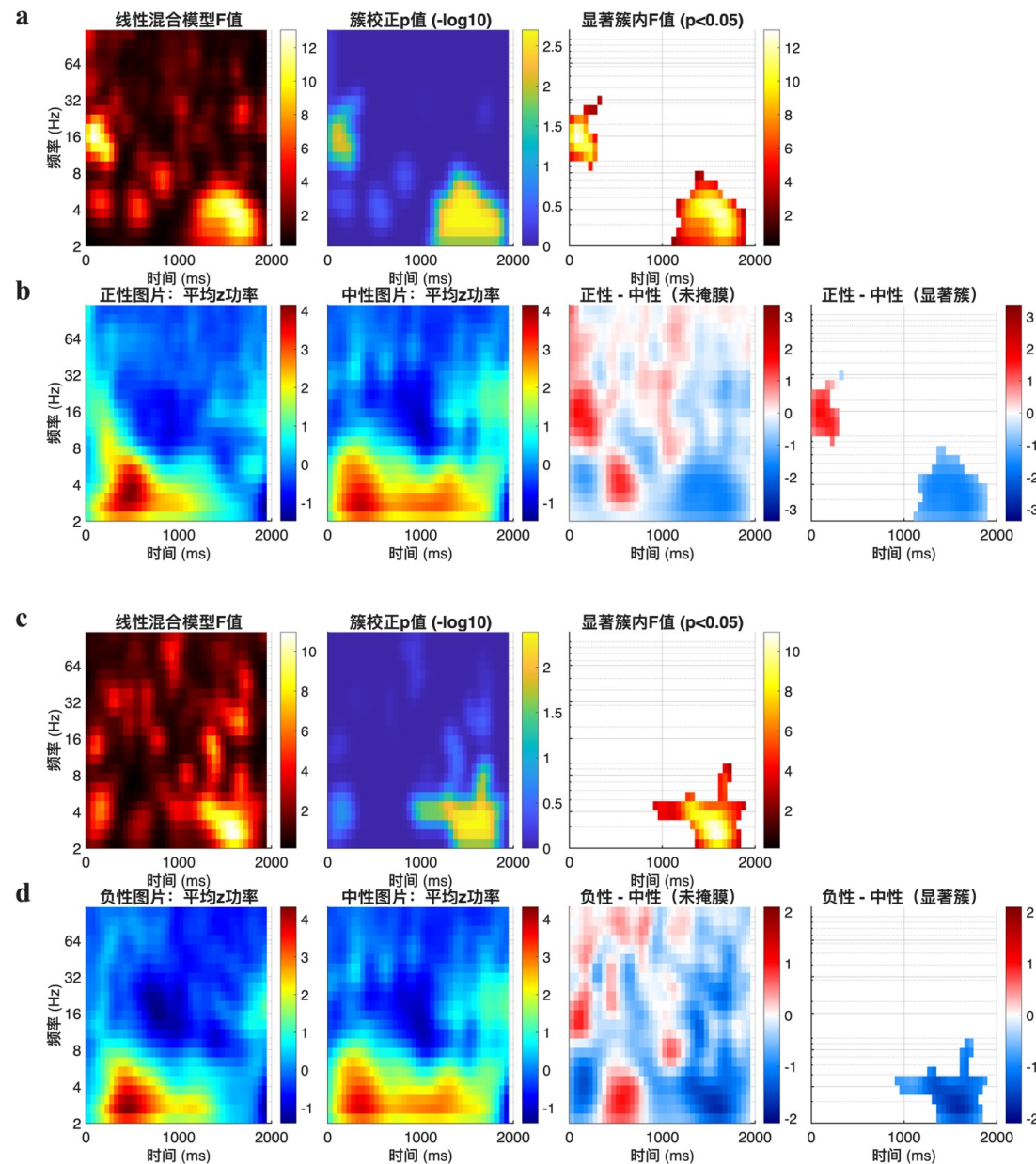
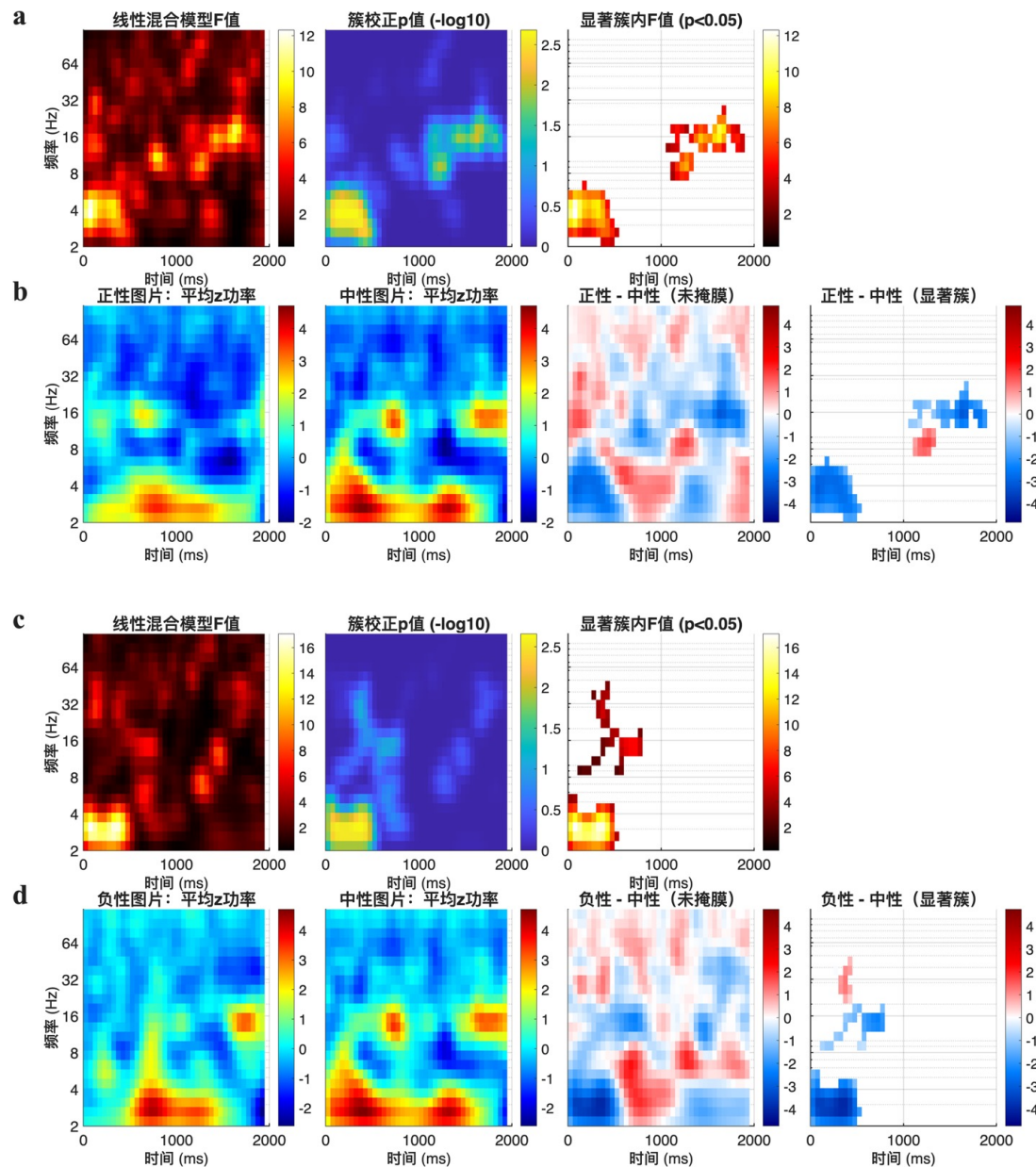
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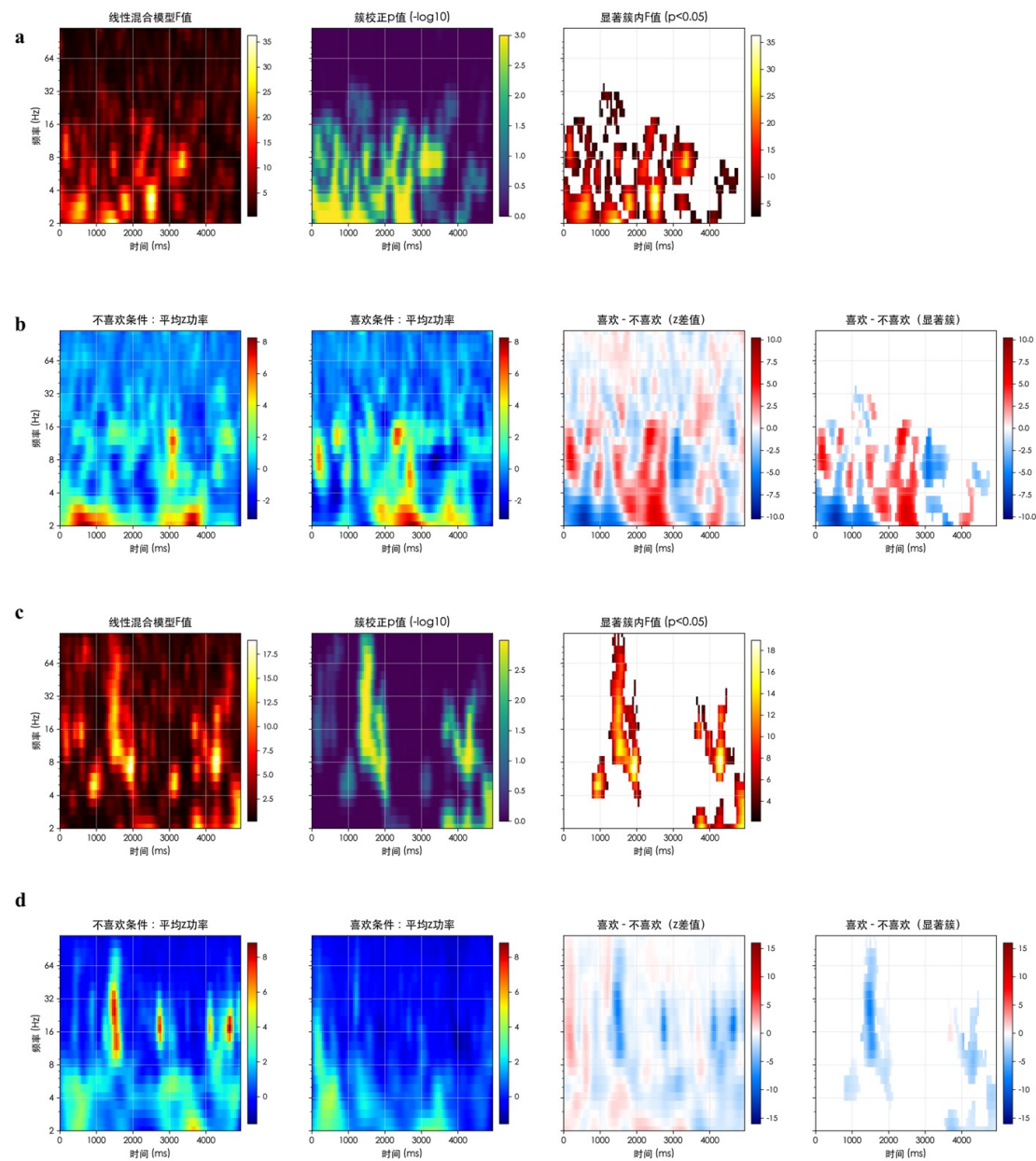












Clip time range: {start_sec:.3f}s to {end_sec:.3f}s. You are rating a 2-second movie clip for AROUSAL (activation intensity), NOT valence (pleasant/unpleasant). From a typical audience perspective, decide whether this 2-second clip is likely to trigger tension, startle, conflict expectation, or rapid emotional escalation. Return ONLY strict JSON with keys: arousal_score (0-100 integer), confidence (0-100 integer), reason_tags (array of short snake_case tags).

```
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